

Optical Modelling of Ocellar Systems

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Background

Insects exhibit rapid and agile flight maneuvers in response to external perturbations. The ocellar visual system plays a key role in these responses. These ‘simple eyes’, which have lower resolution but faster processing when compared to the compound eyes, have been hypothesized to focus on yaw, pitch, and roll detection. We can evaluate optical properties of ocelli in order to improve engineered flight systems, guidance, and control.

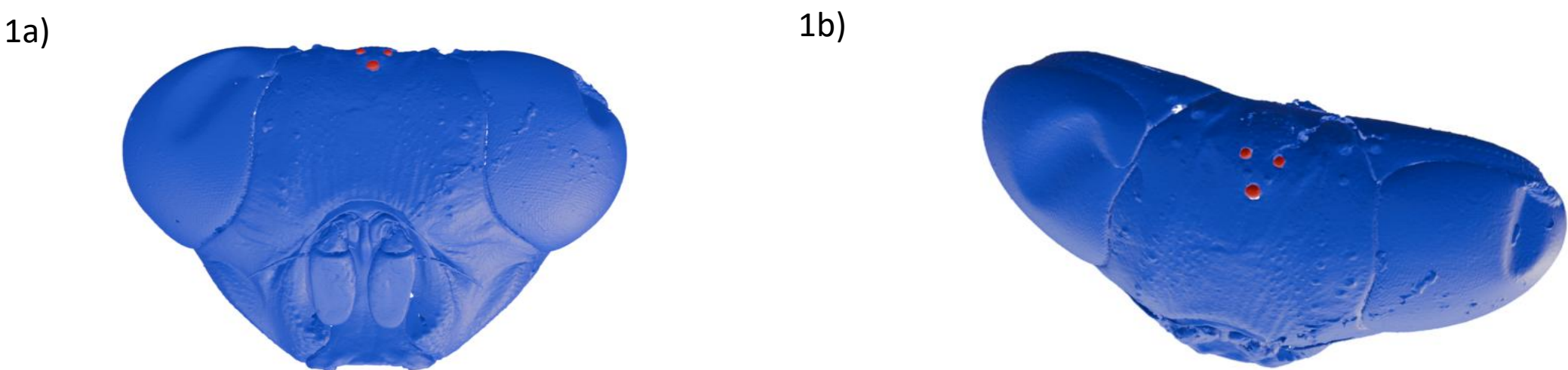


Figure 1) X-Ray micro-computed tomography data of a fly head (blue) with ocelli highlighted in red in two positions. 1a) anterior view; 1b) dorsal view.

Methods

A novel optical ray tracer was developed that recursively casts rays through 3D scans of ocelli to simulate light propagation through the lens system. Proximal and distal surfaces were defined, with rays entering only through the proximal surface and exiting through the distal surface. Rays interact via either total internal reflection or refraction. The pipeline applies variable refractive indices and separate surface interactions to replicate real biological conditions. The ray tracer calculated focal distance and angular acceptance of ocellar systems.

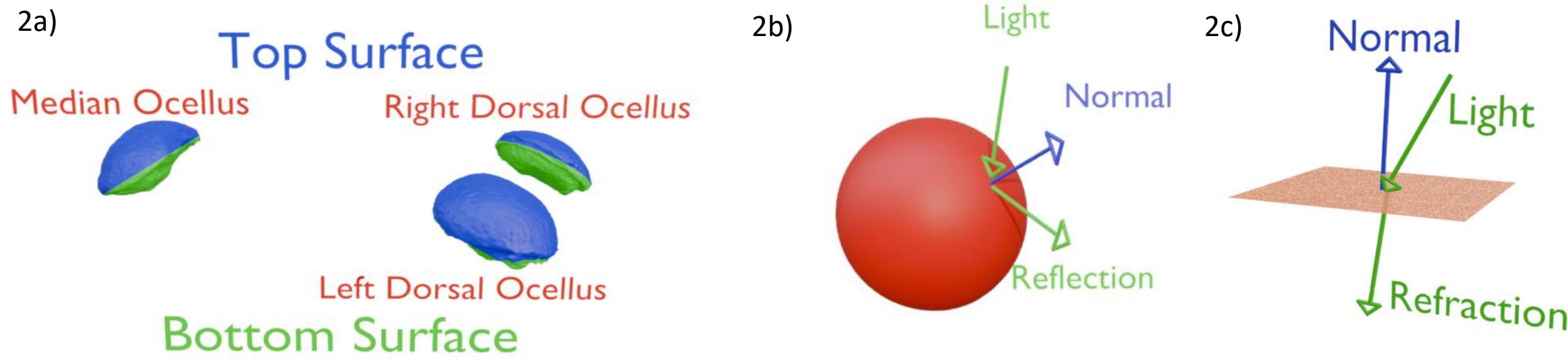


Figure 2) Blender illustrations highlighting key aspects of the optical ray tracer and Snell's Law.

Results

Focal Distance:

Generated stereolithography files of ocellar lens systems were passed through the pipeline to calculate intersection points of rays as they exit the distal surface of an ocellus. These intersection were used to calculate focal distance to estimate focal distances given refractive properties.

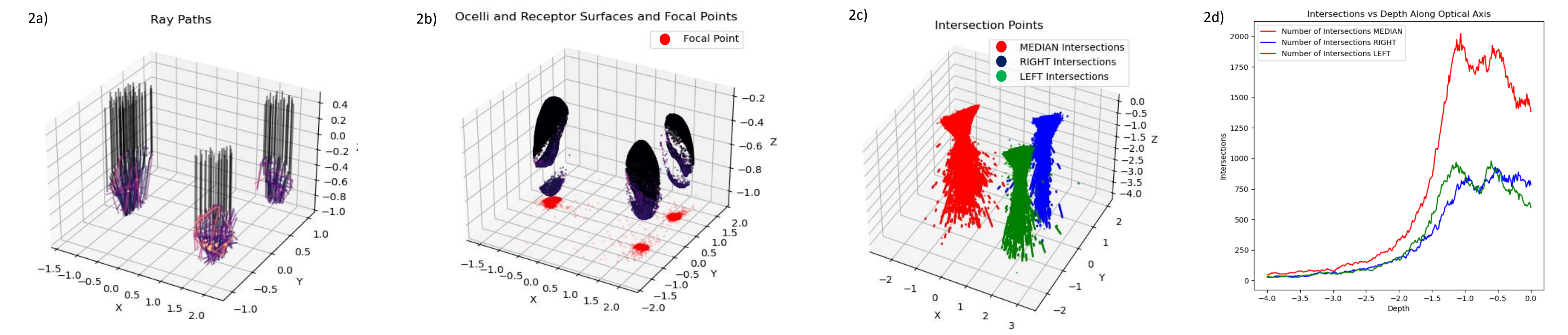


Figure 2) Collimated light cast down along the optical axis onto the ocellar system. 2a) shows the paths that the rays take as they interact with the ocelli, with black showing the first interaction and higher colors showing higher orders of interaction. 2b) shows a point cloud of all the points where rays interact with ocelli with the same color scheme as ray paths. The red dots are the points along the rays at the focal depth. 2c) is a point cloud of intersections (every rays intersection with every other point pairwise computed) for each ocellus. 2d) shows the number of intersections along the optical axis for each ocellus.

Angular Acceptance:

We ran multiple simulations in which the angle of collimated light was varied to calculate the angles of light an ocellar system can detect, given any photoreceptor location and geometry.

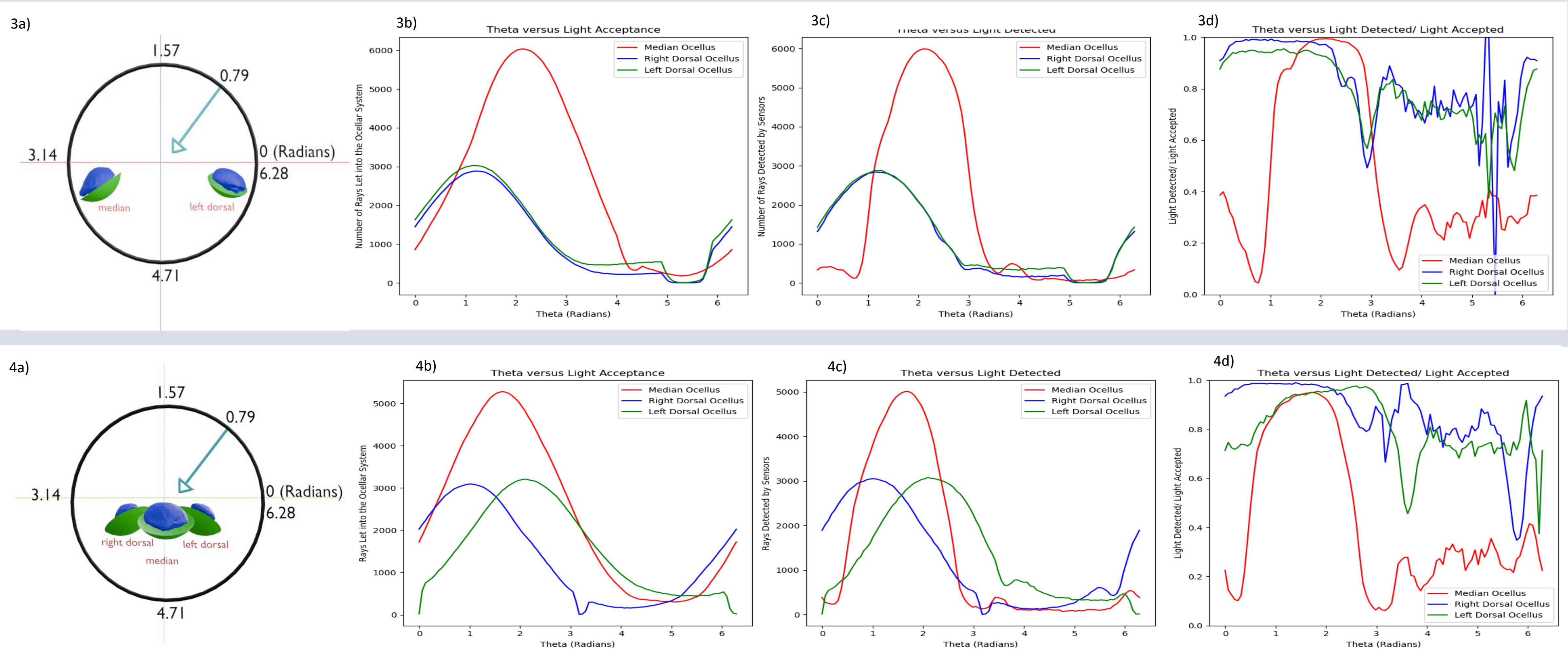


Figure 3 and 4 show angular acceptance as collimated light of different directions are passed through the optical ray tracer. Figures 3a) and 4a) illustrate the plane that the direction of light is changing on and the orientation of the ocellar system. blue being the ocelli and green being the photoreceptor area.. Fig 3b) and 4b) show the number of rays that enter the system, if they hit the top surface of an ocellus. Figures 3c) and 4c) show the number of rays that make it through the system for a given theta. 3d) and 4d) illustrate the ratio between rays detected and the rays that get into the system.

Future Work

Our next steps for this project will be evaluating image quality of rays cast onto photoreceptors by assigning grid values to collimated rays. This would enable researchers to visually ascertain image distortion.

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